# usb_joystick_routed

Design Report

Rev 1 | 2026-10-18 | jlcpcb-tier1

kicad-tools 0.23.0

## Board Summary

| Property | Value |
|---|---|
| Layers | 4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu) |
| Footprints | 38 (37 SMD, 1 THT, 0 other) |
| Nets | 27 |
| Traces | 1080 segments |
| Vias | 105 |
| Board Size | 80.0 x 60.0 mm |

# Stackup

| Layer | Type | Thickness (mm) | Material |
|---|---|---|---|
| F.Cu | copper | 0.035 | -- |
| dielectric 1 | prepreg | 0.2104 | 7628 |
| In1.Cu | copper | 0.0152 | -- |
| dielectric 2 | core | 1.065 | FR4 |
| In2.Cu | copper | 0.0152 | -- |
| dielectric 3 | prepreg | 0.2104 | 7628 |
| B.Cu | copper | 0.035 | -- |

# Design Overview

## Theory of Operation

USB joystick controller — ATmega32U4

USB bus powered, 8MHz crystal, AVR ISP programming

GPIO and ADC map is fixed to Microchip TQFP-44 pinout

## Communication Interfaces

| Protocol | Signals |
|---|---|
| SPI | ISP_MISO, ISP_MOSI, ISP_SCK |
| USB | USB_D+, USB_D-, VBUS |

# Assembly Notes

1 fine-pitch component

- **Fine-pitch components**: 1 (U1)

# Hand-Solder (THT) Components

The following 1 through-hole component is **excluded from the SMT pick-and-place file** and must be hand-soldered (or wave/selective-soldered) after SMT assembly. It appears in the BOM for sourcing.

| Value | Package | Qty | References |
|---|---|---|---|
| USB4085-GF-A | USB_C_Receptacle_GCT_USB4085 | 1 | J1 |

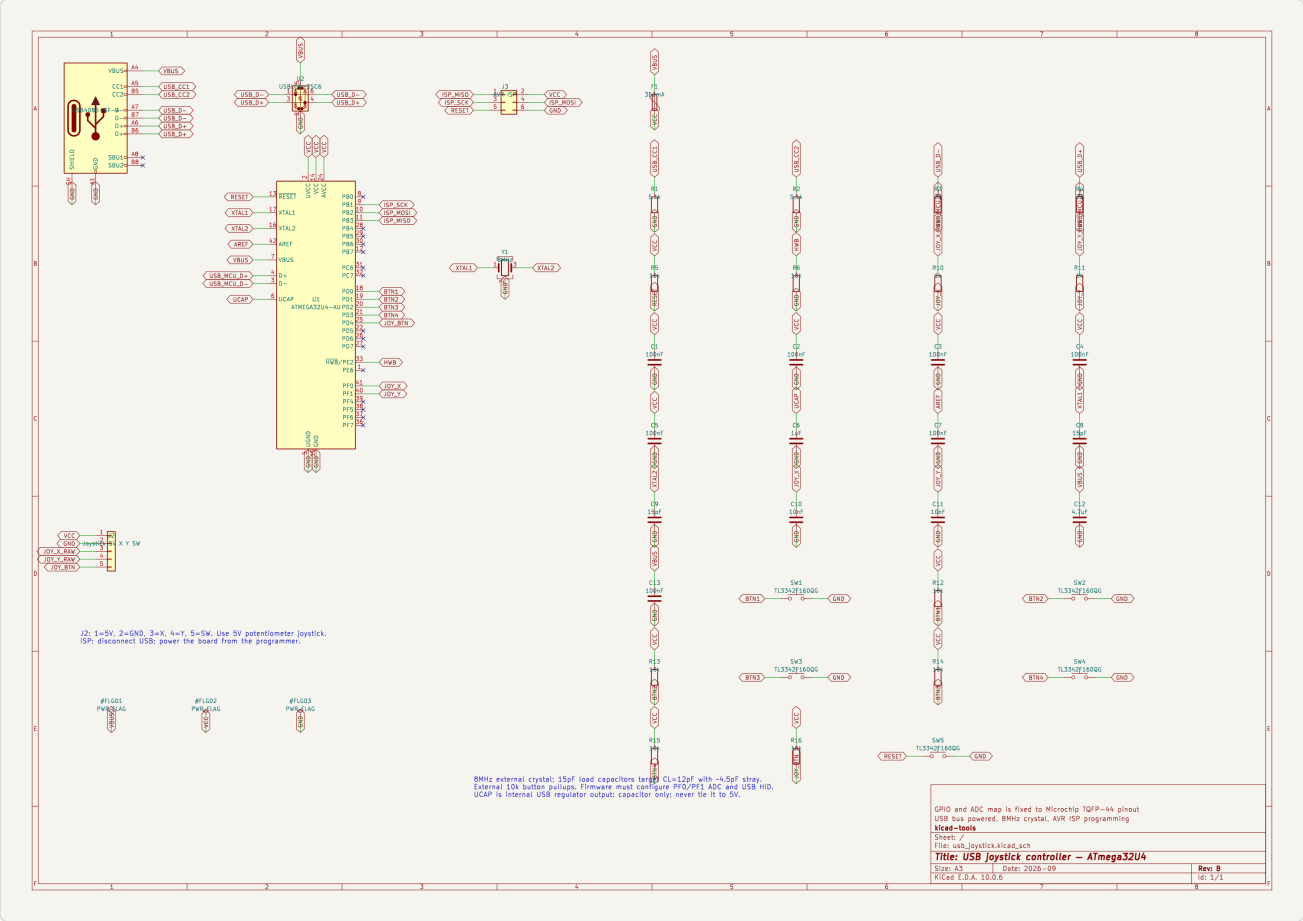
# ERC Status

| Metric | Count |
|---|---|
| Errors | 0 |
| Warnings | 0 |

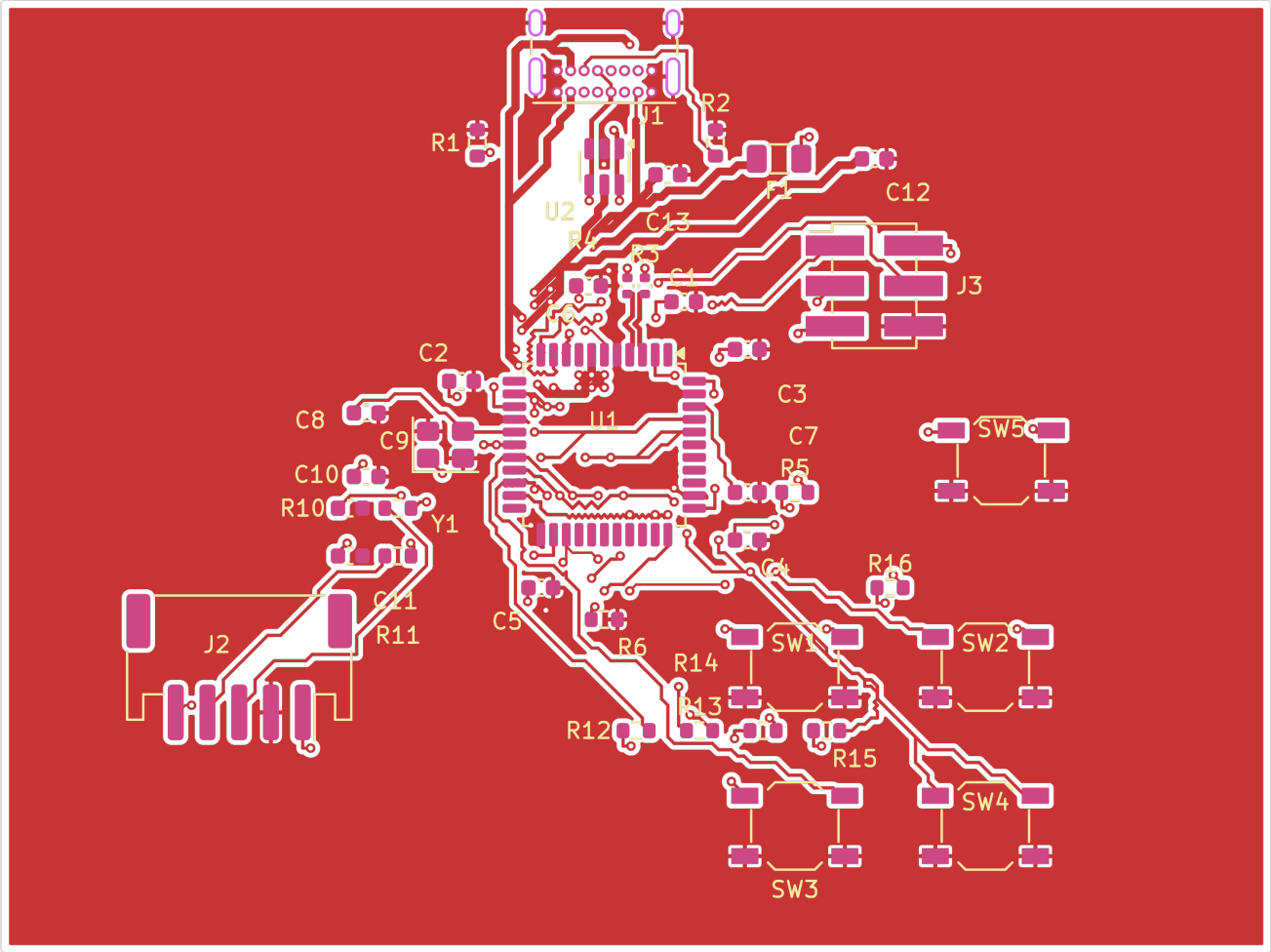
**Status**: SKIPPED -- ERC skipped by user request

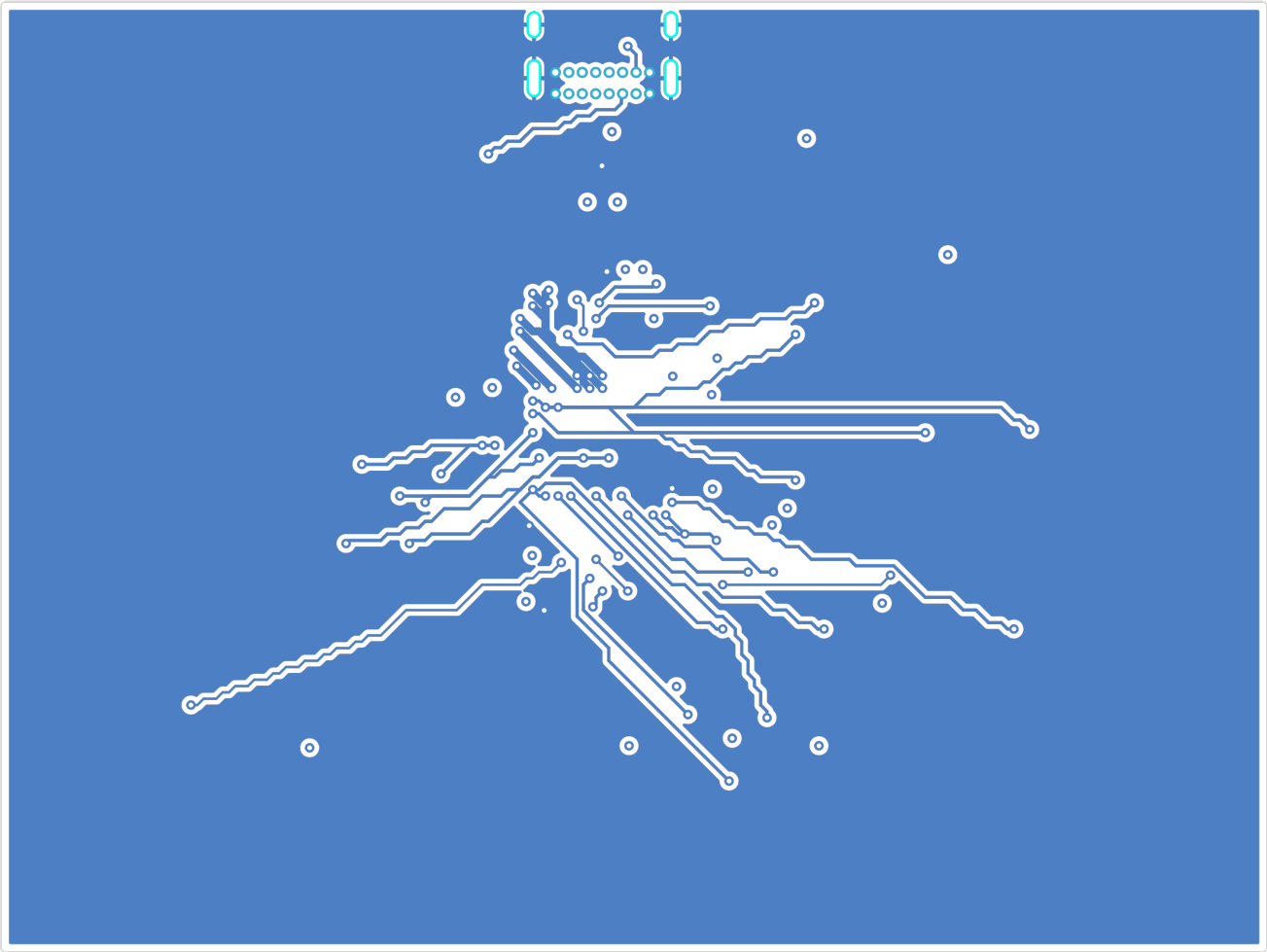
# Schematic Overview

## Schematic: usb_joystick

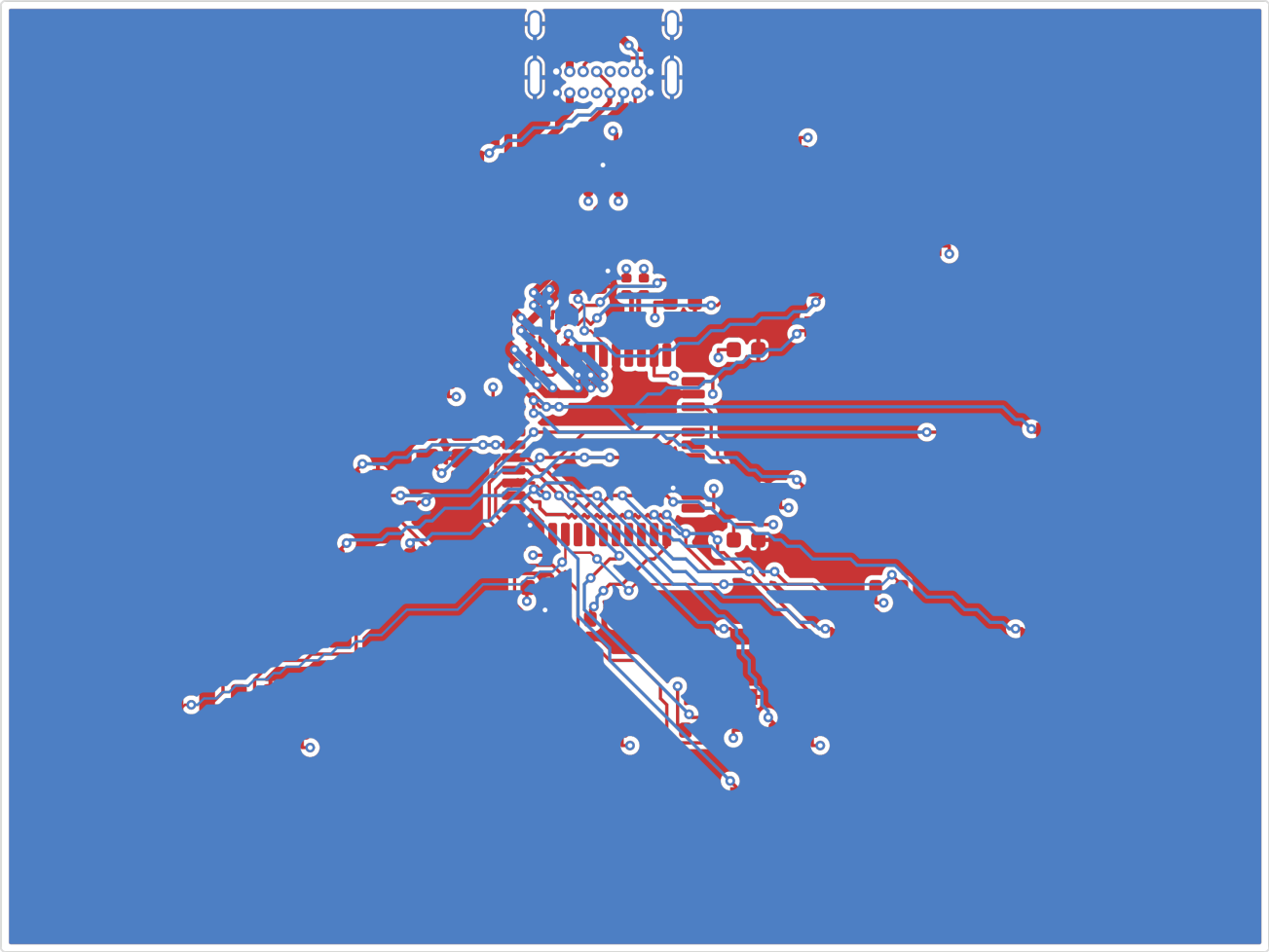


# PCB Layout

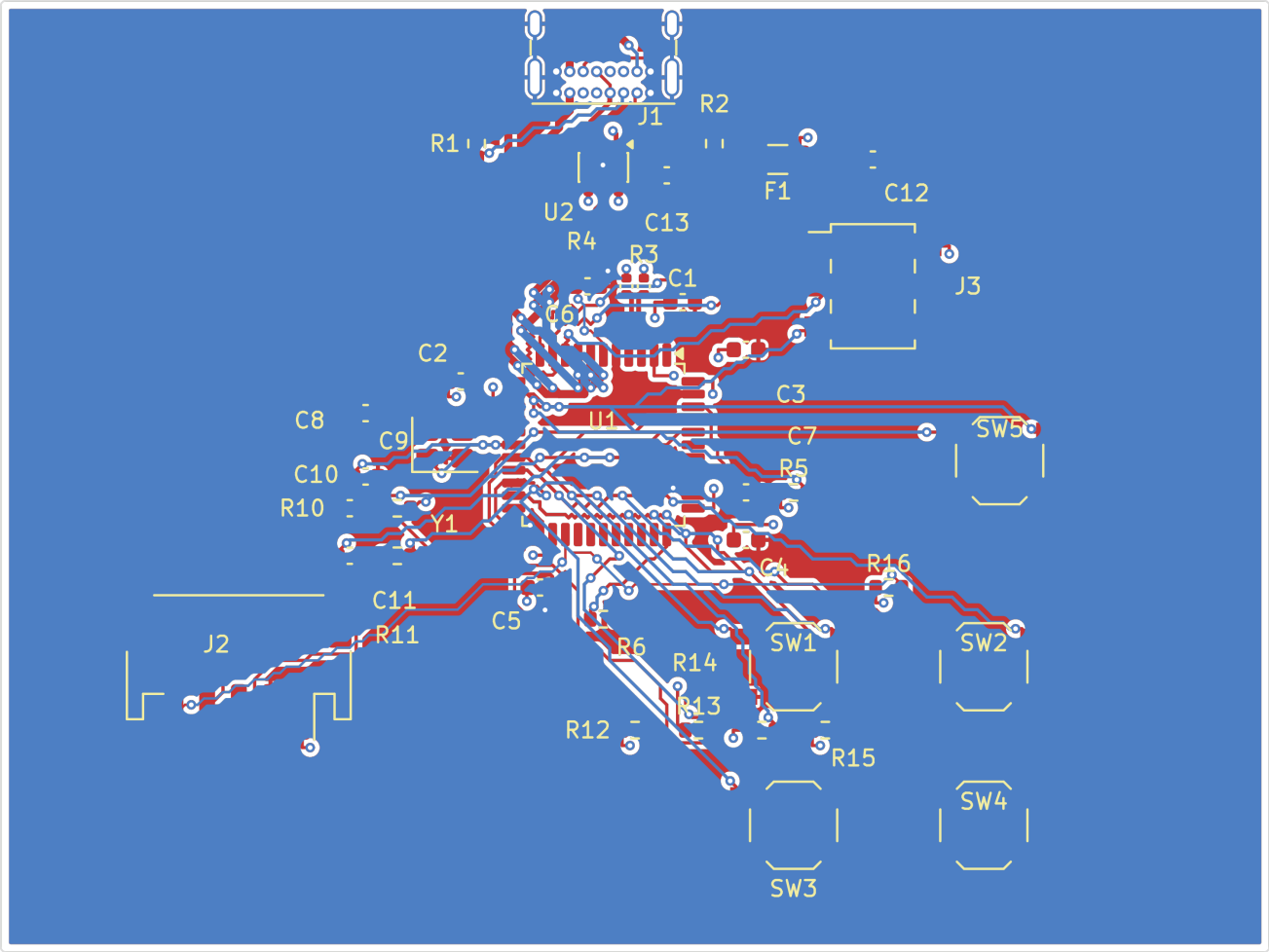

R1
J1
R2
U2
C13
C12
J3
C1
C2
C3
C8
C9
U1
C7
SW5
C10
R10
Y1
R5
C4
R16
C11
R11
J2
C5
R6
SW1
SW2
R14
R13
R12
R15
SW4
SW3



## Copper

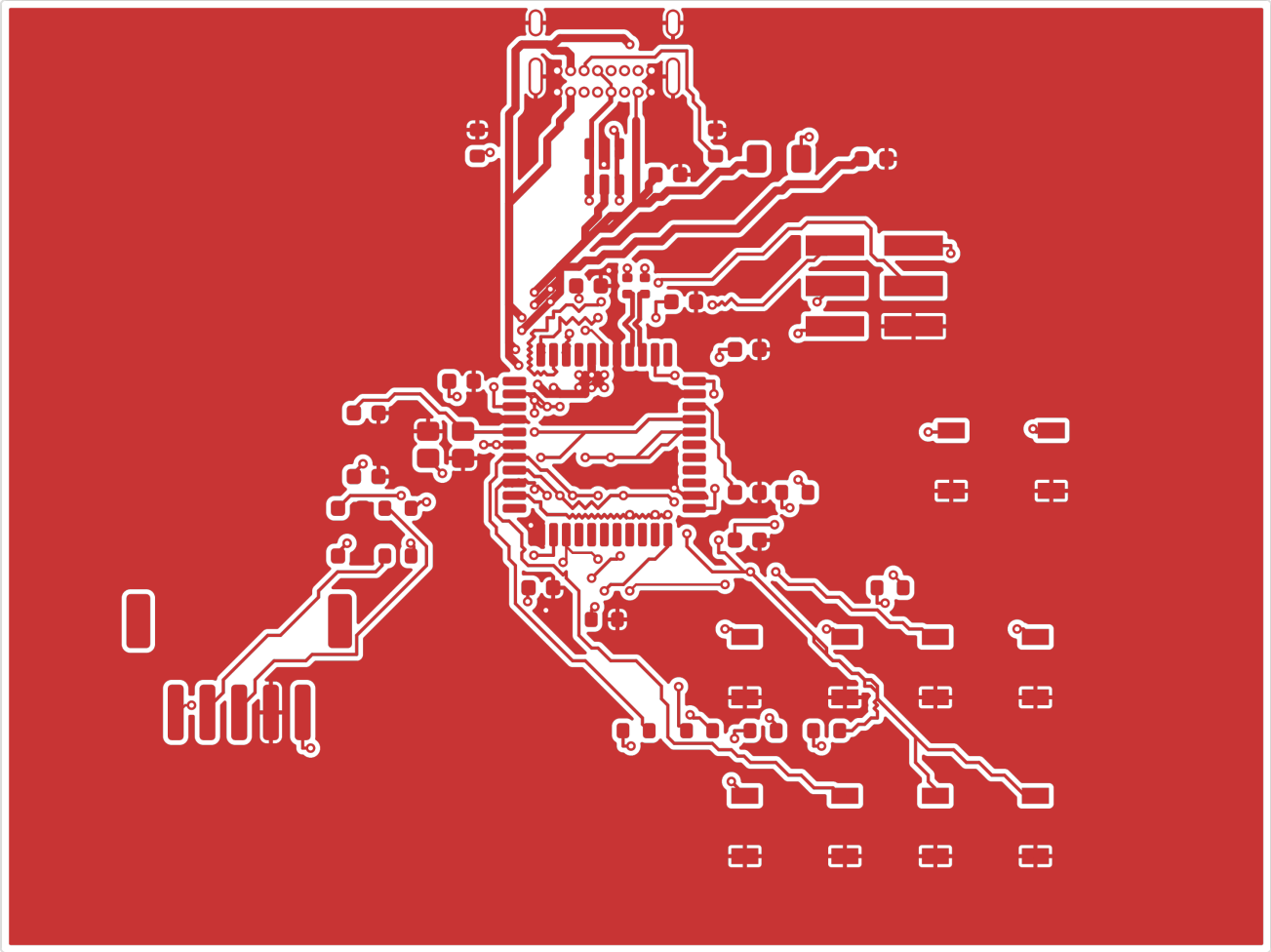


## Assembly

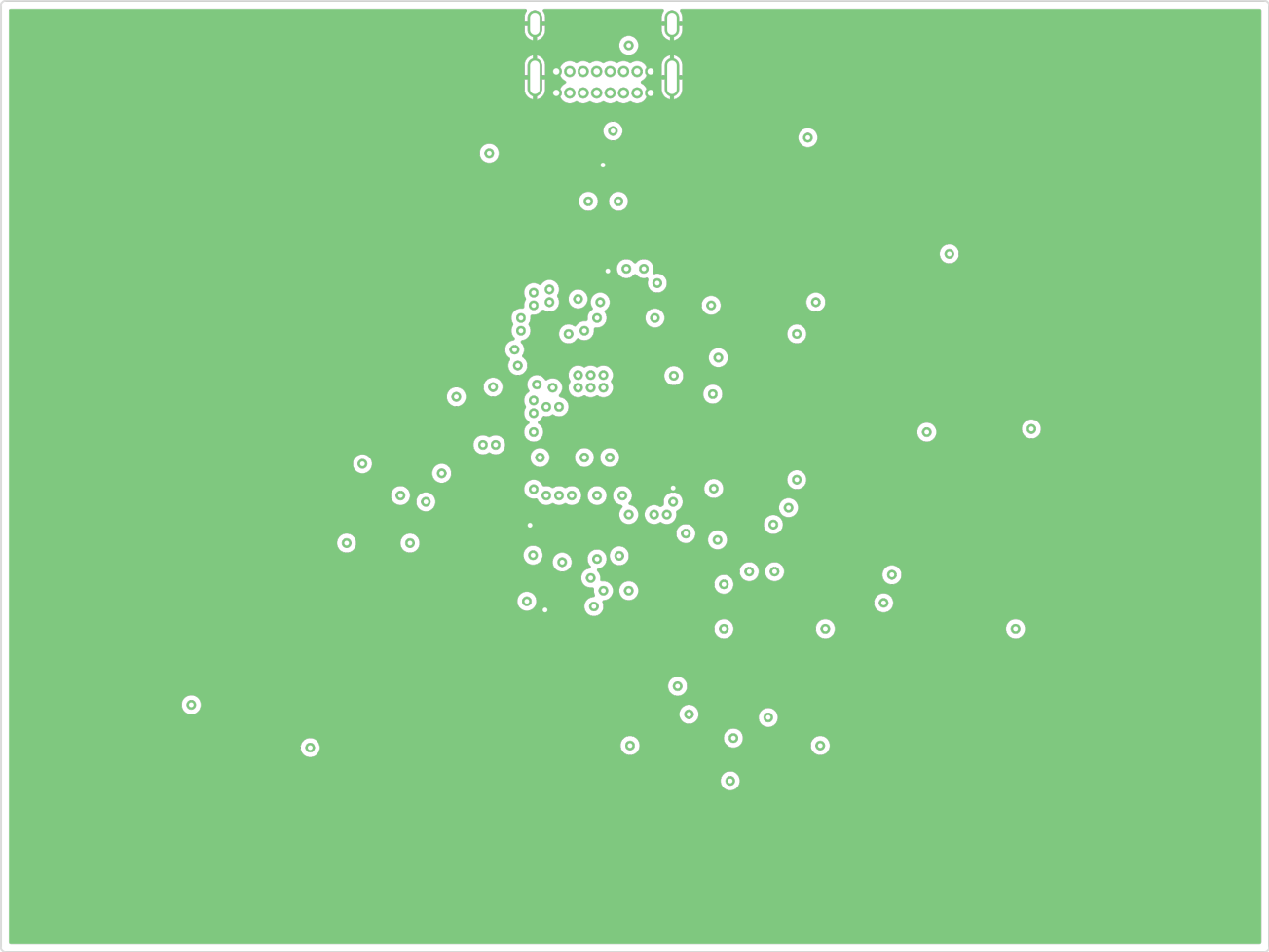

J1
R2
R1
F1
C12
U2
C13
R4
R3
C1
J3
C6
C2
C3
C8
C9
U1
C7
SW5
C10
R5
R10
Y1
R16
C11
C5
R11
J2
R6
SW1
SW2
R14
R13
R12
R15
SW4
SW3

# Copper Layers

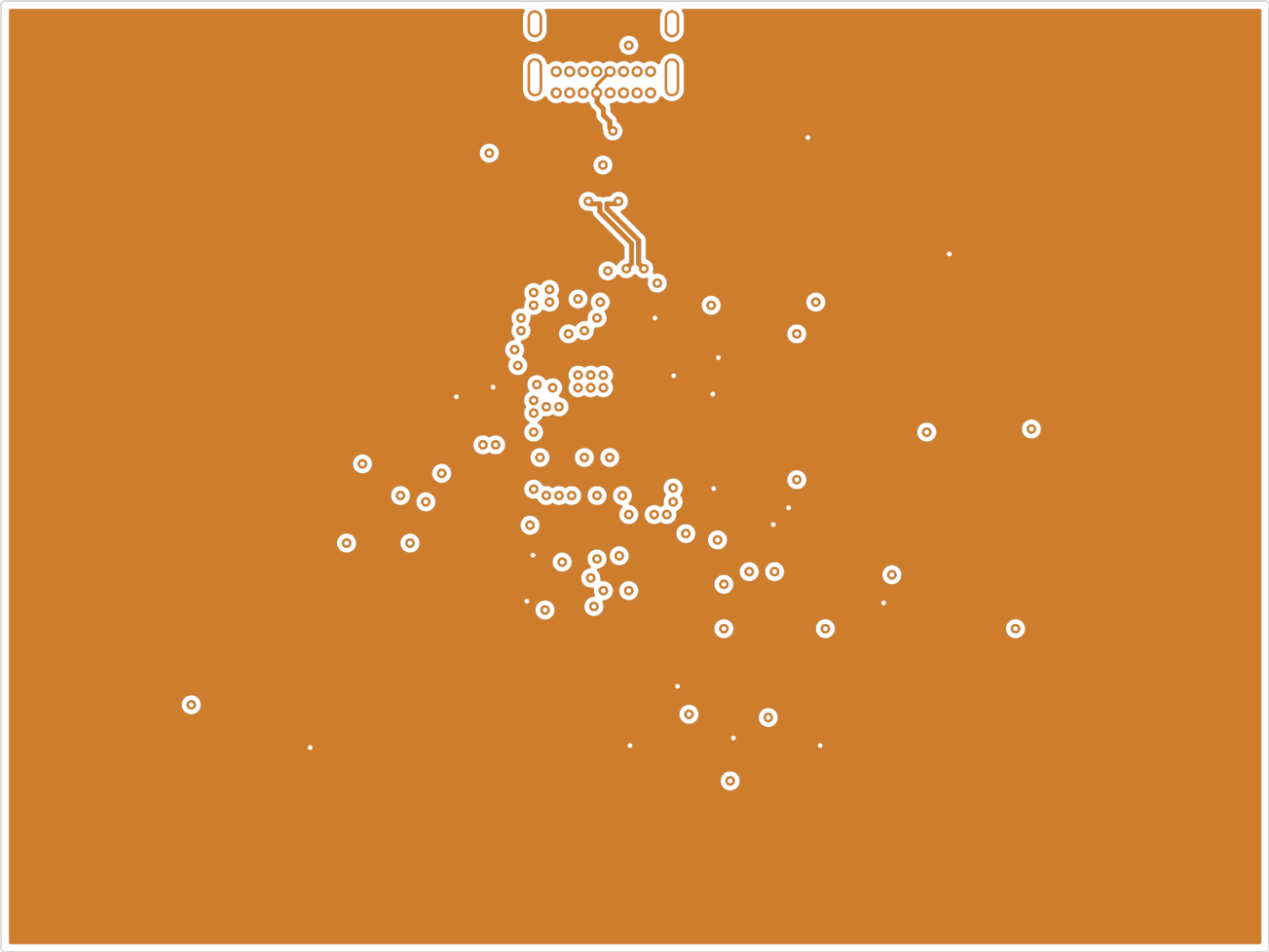
## F.Cu



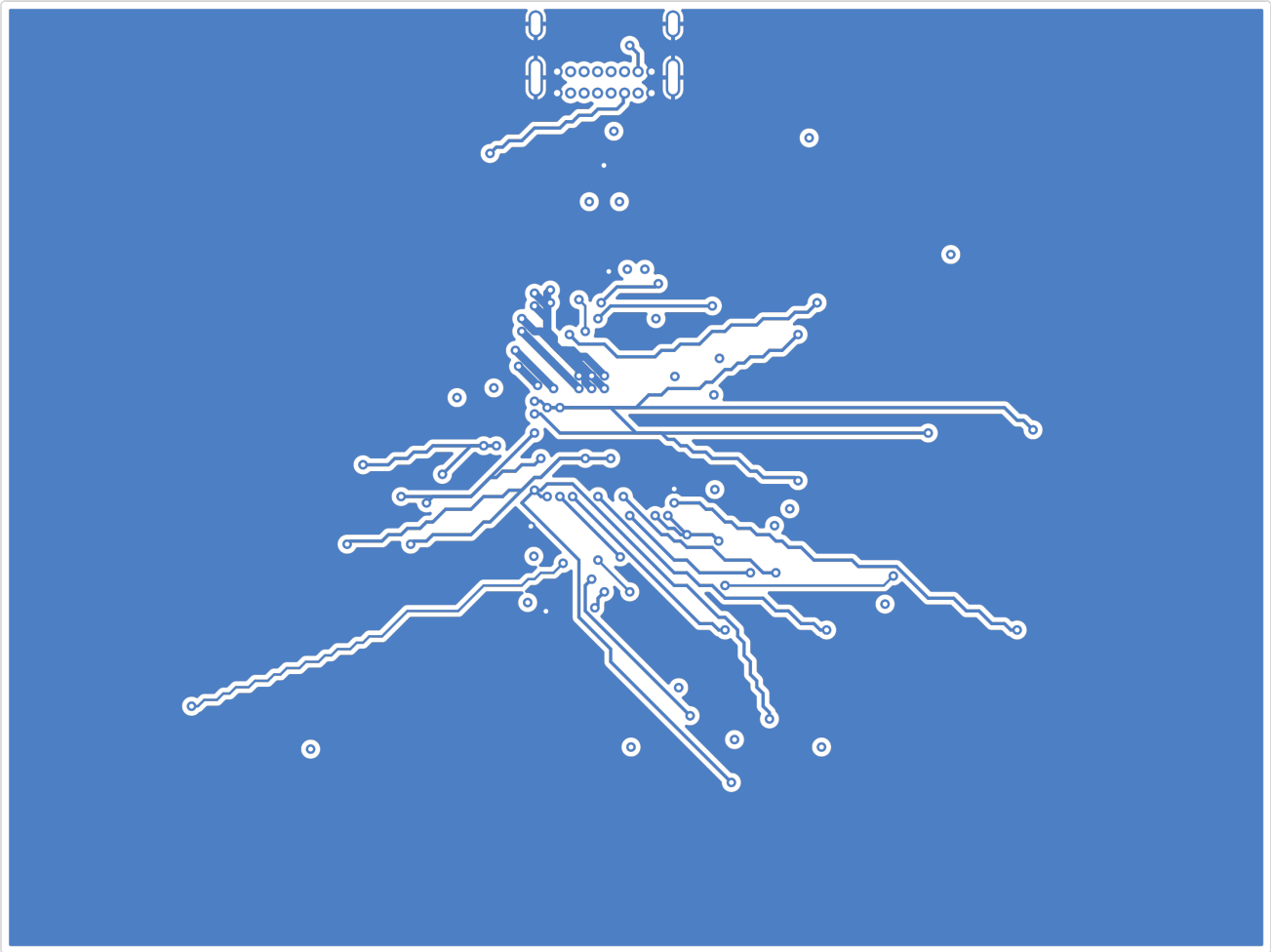
## In1.Cu



## In2.Cu



## B.Cu



usb_joystick_routed

# Bill of Materials

| Value | Package | Qty | References | MPN | LCSC |
|---|---|---|---|---|---|
| 100nF | C_0603_1608 Metric | 7 | C1, C2, C3, C4, C5, C7, C13 | GRM188R71C 104KA01D | C45000 |
| 10nF | C_0603_1608 Metric | 2 | C10, C11 | GRM188R71H 103KA01D | C77053 |
| 15pF | C_0603_1608 Metric | 2 | C8, C9 | GRM1885C1H 150JA01D | C71651 |
| 1uF | C_0603_1608 Metric | 1 | C6 | GRM188R71A 105KA61D | C97888 |
| 4.7uF | C_0603_1608 Metric | 1 | C12 | GRM188R61A 475KE15D | C71633 |
| 350mA | Fuse_1206_32 16Metric | 1 | F1 | 1206L035/16Y R | C126817 |
| AVR ISP | Samtec_TSM-103-01-T-DV | 1 | J3 | TSM-103-01-T-DV-P-TR | C21285200 |
| Joystick 5V X Y SW | JST_PH_S5B-PH-SM4-TB_1x05-1MP_P2.00mm_Horizontal | 1 | J2 | S5B-PH-SM4-TB(LF)(SN) | C265104 |
| USB4085-GF-A | USB_C_Receptacle_GCT_USB4085 | 1 | J1 | USB4085-GF-A | C7095263 |
| 10k | R_0603_1608 Metric | 7 | R5, R6, R12, R13, R14, R15, R16 | RC0603FR-07 10KL | C98220 |
| 1k | R_0603_1608 Metric | 2 | R10, R11 | RC0603FR-07 1KL | C22548 |
| 22 | R_0402_1005 Metric | 2 | R3, R4 | RC0402FR-07 22RL | C114765 |
| 5.1k | R_0603_1608 Metric | 2 | R1, R2 | RC0603FR-07 5K1L | C105580 |
| TL3342F160Q G | SW_SPST_TL3342 | 5 | SW1, SW2, SW3, SW4, SW5 | TL3342F160Q G | C2886898 |

| Value | Package | Qty | References | MPN | LCSC |
|---|---|---|---|---|---|
| ATMEGA32U4-AU | TQFP-44_10x10mm_P0.8mm | 1 | U1 | ATMEGA32U4-AU | C44854 |
| USBLC6-2SC6 | SOT-23-6 | 1 | U2 | USBLC6-2SC6 | C7519 |
| 8MHz | Crystal_SMD_3225-4Pin_3.2x2.5mm | 1 | Y1 | ECS-80-12-33-JGN-TR | C2442665 |

# DRC Status

| Metric | Count |
|---|---|
| Errors | 0 |
| Warnings | 15 |
| Blocking | 0 |

**Status**: PASS

## Violations by Type

| Violation Type | Count |
|---|---|
| hole_to_hole_clearance | 14 |
| diffpair_routing_continuity | 1 |
| diffpair_uncoupled_length | 1 |

# Manufacturing Readiness

**Verdict**: WARNING

## Action Items

- **[OPTIONAL]** Refill zones to verify 2 zone-connected nets, then re-run kct audit
- **[OPTIONAL]** Review 15 DRC warnings

# Routing Status

| Metric | Value |
|---|---|
| Signal Net Completion | 100.0% (25/25) |
| Overall Completion | 100.0% |
| Complete Nets | 27 / 27 |
| Zone-Connected Nets | 2 |
| Incomplete Nets | 0 |
| Unconnected Pads | 0 |

## Zone-Connected Nets

- GND
- VCC

# Cost Estimate

| Metric | Per Board (estimated) |
|---|---|
| PCB Fabrication | ~2.96 USD |
| Components (estimated) | ~2.11 USD |
| Assembly (estimated) | ~2.16 USD |
| **Total (estimated)** | **~7.23 USD** |
| Batch Quantity | 5 |
| Batch Total (estimated) | ~36.14 USD |